

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0261

Meadowbrook 100 kV
July 20, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0261. The Interconnection Customer proposes a 259.8 MW wind facility interconnecting at the Meadowbrook 100 kV substation in the Indiana area. The request was received on August 13, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 200 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$87,564,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Madison / Pendleton 100 KV is overloaded to 339.1% of its 85 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 / Southeast Vicksburg 500 KV. The project contributes 100.0% DFAX (259.8 MW) to this overload.

The South Haven / Anderson 100 KV is overloaded to 265.9% of its 113 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 / Southeast Vicksburg 500 KV line. The project contributes 100.0% DFAX (259.8 MW) to this overload.

Madison / Meadowbrook 100 KV: 250.9% loading (125 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 / Southeast Vicksburg 500 KV; DFAX = 100.0%.

The S Haven / Rmec Substation 100 KV is overloaded to 214.7% of its 120 MVA emergency rating for the loss of the South Haven / Napavine 100 KV. The project contributes 100.0% DFAX (259.8 MW) to this overload.

The Anderson / Pendleton 100 KV L is overloaded to 211.4% of its 125 MVA emergency rating for the loss of the Southeast Vicksburg / Allegheny Energy Units 3 4 & 5 500 KV. The project contributes 100.0% DFAX (259.8 MW) to this overload.

The S Haven / Napavine 100 KV is overloaded to 199.4% of its 129 MVA emergency rating for the loss of the Rmec / S Haven 100 KV. The project contributes 100.0% DFAX (259.8 MW) to this overload.

For the loss of Rmec / S Haven Substation 100 KV line, the Napavine / Murdock 100 KV L loads to 185.9% of its 133 MVA rating. Project DFAX: 100.0%.

The Hansel / Butler Municipal Power Plant 100 KV is overloaded to 133.9% of its 128 MVA emergency rating for the loss of the E Methuen 500 kV/230 kV Transformer. The project contributes 5.4% DFAX (14.1 MW) to this overload.

The Richard Lake / Savona 100 KV is overloaded to 133.1% of its 110 MVA emergency rating for the loss of the E Methuen 500-230 kV Transformer. The project contributes 5.4% DFAX (14.1 MW) to this overload.

Butler Municipal Power Plant / Savona 100 KV: 131.6% loading (121 MVA) following the outage of the East Methuen Substation 500 kV/230 kV XFMR; DFAX = 5.4%.

The Christopher / Rmec 100 KV Line is overloaded to 131.1% of its 124 MVA emergency rating for the loss of the South Haven / Napavine 100 KV. The project contributes 73.0% DFAX (189.5 MW) to this overload.

For the loss of E Methuen 500/230 kV Transformer Bank, the Hansel / Thousand Springs 100 KV loads to 126.2% of its 142 MVA rating. Project DFAX: 5.4%.

For the loss of Montpelier Electric Generating Sta / Wabash 500 KV, the Montpelier Electric Generating Sta Substation / Dooville 230 KV L loads to 115.2% of its 562 MVA rating. Project DFAX: 8.9%.

For the loss of Swan Valley / Trego 500 KV L, the Timber Switch / Muskogee 230 KV loads to 111.3% of its 417 MVA rating. Project DFAX: 6.3%.

Richard Lake / Carbondale Nw 100 KV: 109.8% loading (116 MVA) following the outage of the E Methuen 500/230kV XFMR; DFAX = 5.4%.

Coronet / Westwood Generation Llc 500 KV: 106.1% loading (2,444 MVA) following the outage of the Acadia / Montour Station 500 KV; DFAX = 11.2%.

Youngwood / Woodland Tap 345 KV line: 104.8% loading (1,494 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 / Southeast Vicksburg 500 KV; DFAX = 6.3%.

Shawville Sta 500/345kV XFMR: 104.7% loading (867 MVA) following the outage of the Carbon Center 500/345kV XFMR; DFAX = 5.4%.

For the loss of Montpelier Electric Generating Station / Wabash 500 KV, the Montpelier Electric Generating Sta 500/230 kV Autotransformer loads to 104.6% of its 804 MVA rating. Project DFAX: 11.2%.

Swan Valley 500/230 kV TX: 104.3% loading (1,041 MVA) following the outage of the Swan Valley / Trego 500 KV; DFAX = 14.6%.

Omni-Hackerry Substation / Norwich Substation 500 KV: 104.0% loading (2,332 MVA) following the outage of the Trego / Beaver 500 KV; DFAX = 25.2%.

For the loss of Carbon Ctr 500/345kV Auto, the Shawville Sta / Blackrock 345 KV loads to 103.5% of its 1,339 MVA rating. Project DFAX: 8.4%.

For the loss of Norwich / Barrett 500 KV line, the Minco / Larson Street 230 KV loads to 102.5% of its 1,056 MVA rating. Project DFAX: 7.5%.

The Allegheny Energy Units 3 4 & 5 345-1 kV Autotransformer is overloaded to 102.1% of its 1,169 MVA emergency rating for the loss of the Hilltop / Piney Creek Project 500 KV L. The project contributes 10.7% DFAX (27.8 MW) to this overload.

For the loss of Hilltop / Piney Creek Project 500 KV, the Allegheny Energy Units 3 4 & 5 500/1 kV Transformer loads to 102.1% of its 1,169 MVA rating. Project DFAX: 10.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Madison / Pendleton 100 KV	Rebuild line with higher-capacity conductor	\$3,104,000	100% (\$3,104,000)
S Haven / Anderson 100 KV	Rebuild line with higher-capacity conductor	\$5,087,000	100% (\$5,087,000)
Madison / Meadowbrook 100 KV	Rebuild line with higher-capacity conductor	\$1,741,000	100% (\$1,741,000)
Rmec / S Haven 100 KV	Rebuild line with higher-capacity conductor	\$6,106,000	100% (\$6,106,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pendleton / Anderson 100 KV	Rebuild line with higher-capacity conductor	\$2,601,000	100% (\$2,601,000)
S Haven / Napavine 100 KV	Rebuild line with higher-capacity conductor	\$5,062,000	100% (\$5,062,000)
Napavine / Murdock 100 KV line	Rebuild line with higher-capacity conductor	\$20,012,000	100% (\$20,012,000)
Hansel / Butler Municipal Power Plant 100 KV	Rebuild line with higher-capacity conductor	\$3,436,000	30% (\$1,045,000)
Richard Lake / Savona Substation 100 KV Line	Rebuild line with higher-capacity conductor	\$10,253,000	30% (\$3,118,000)
Butler Municipal Power Plant / Savona 100 KV	Rebuild line with higher-capacity conductor	\$1,387,000	30% (\$422,000)
Rmec / Christopher 100 KV	Rebuild line with higher-capacity conductor	\$3,620,000	100% (\$3,620,000)
Hansel / Thousand Springs 100 KV	Rebuild line with higher-capacity conductor	\$6,220,000	30% (\$1,892,000)
Dooville / Montpelier Electric Generating Station 230 KV	Rebuild line with higher-capacity conductor	\$45,713,000	12% (\$5,662,000)
Timber Switch / Muskogee 230 KV	Rebuild line with higher-capacity conductor	\$19,993,000	9% (\$1,798,000)
Richard Lake Substation / Carbondale Nw 100 KV L	Reconductor line	\$1,618,000	30% (\$492,000)
Coronet / Westwood Generation Llc 500 KV	Reconductor line	\$70,361,000	7% (\$4,617,000)
Youngwood / Woodland Tap 345 KV line	Reconductor line	\$13,035,000	6% (\$844,000)
Shawville Station 500/345 kV XFMR	Replace transformer with higher-rated unit	\$27,370,000	28% (\$7,537,000)
Montpelier Electric Generating Sta 500/230 kV XFMR	Replace transformer with higher-rated unit	\$29,209,000	7% (\$2,086,000)
Swan Vly 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,977,000	10% (\$2,918,000)
Omni-Hackerry Substation / Norwich Substation 500 KV	Reconductor line	\$35,677,000	6% (\$2,318,000)
Shawville Sta / Blackrock 345 KV	Reconductor line	\$24,735,000	6% (\$1,611,000)
Minco / Larson St 230 KV	Reconductor line	\$2,797,000	8% (\$233,000)
Allegheny Enrgy Units 3 4 & 5 345-1 kV Auto	Replace transformer with higher-rated unit	\$20,548,000	6% (\$1,326,000)
Allegheny Energy Units 3 4 & 5 500/1 kV Transformer	Replace transformer with higher-rated unit	\$35,838,000	6% (\$2,312,000)
	Total allocated to project		\$87,564,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.